INORGANIC CUME
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1. Stepwise formation constants for the complex of NH3 with $[Cu(OH_2)_6]^{2+}$ (aq) are log K_{f1} = 4.15, log K_{f2} = 3.50, log K_{f3} = 2.89, log K_{f4} = 2.13 and log K_{f5} = -0.52. Rationalize why K_{f5} is so different.

2. Octohedral complexes of metals with higher oxidation states or of metals of the second and third row are less labile than those of lower oxidation state and d metals of the first row. Account for this observation on the basis of a dissociative rate-determining step.

3. The rate constant for the formation of $[CoX(NH_3)_5]^{2+}$ from $[Co(NH_3)_5(OH_2)]^{3+}$ for X = Cl^-, Br^-, N_3^- and SCN^- differ by no more than a factor of two. What does this suggest about the mechanism of substitution?

4. Write out the inner- and outer-sphere pathways for the reduction of $[Co(NH_3)_5N_3]^{2+}$ with V^{2+}(aq). What experimental data might be used to distinguish the two pathways?

5. The rate of loss of chlorobenzene, PhCl, from [W(CO)4LPhCl] increases with increases in the cone angle of ligand L. What does this observation suggest about the mechanism of reaction?